

Absolute Gravity Measurements at Maitri, Antarctica

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ABSTRACT

During the 23rd Indian Antarctic expedition, absolute gravity measurements (AG) are made at Maitri, Indian station in Antarctica. This was first AG in the Dronning Maud Land, Antarctica coincidentally just before the Finnish Geodetic Institute (FGI) measurements at Novolazarevskaya, Russian station near Maitri. Measurements are initiated with two major objectives: (i) establishing reference gravity station for future gravimetric surveys, (ii) for studying temporal gravity changes related to Antarctic ice mass loss.

INTRODUCTION

Earth's gravity field is responsive to the changes in the vertical position and mass distribution and therefore, study of gravity field can be utilised to decipher geodynamic phenomenon, which produces mass and vertical changes. Gravity measurements in Antarctica started about 50 years back. Gravity measurements during the Indian expedition was carried out during Indian Antarctic Expedition (IAE) using relative gravity meter and tied it to reference gravity station at Goa. Such measurements can be used for regional studies with a survey accuracy of few mGal. These data cannot be merged with another set of data due to the uncertainty of their reference station and accuracies. Further more these data can also be not used for any temporal studies. Antarctic land mass has very few absolute gravity values and none of them were near to Indian station (Maitri) (Figure 1). Therefore, Absolute reference gravity station was necessity in this part of Antarctica. Development of new generation of gravity meters can measure absolute gravity field to the precision of few microGal (Niebauer et al., 1995). These instruments are commonly utilised to measure changes in the gravity field to understand the redistribution of mass and vertical crustal motions caused due to several geodynamic processes such as earthquakes, plate convergence, isostatic rebound, etc. Study of mass and elevation change in

the Polar Regions are initiated recently (Wahr et al, 2000; Makinen et al., 2007). National Geophysical Research Institute (NGRI) in collaboration with Department of Ocean Development (DoD) has acquired first Absolute Gravity Meter (AG) meter in country with objective to establish reference gravity station in India and Antarctica.

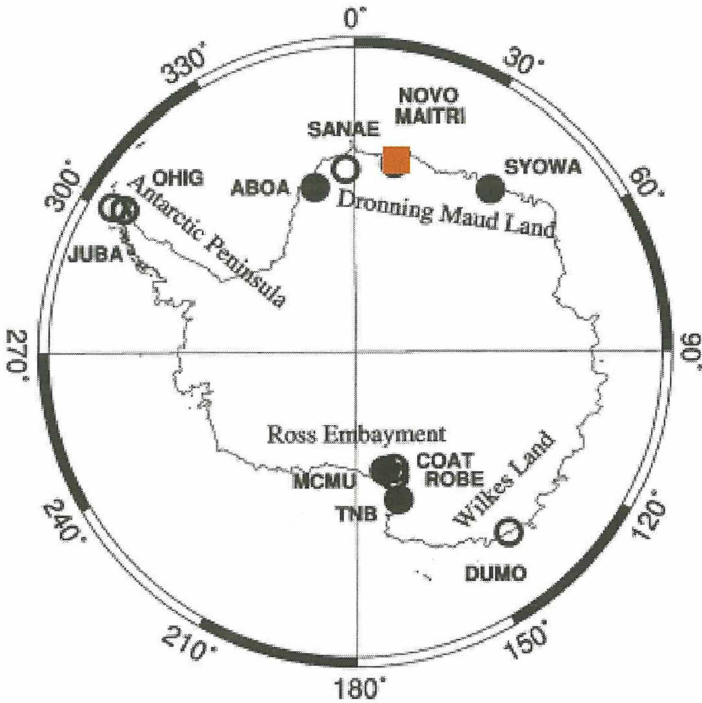


Fig. 1: Location of Absolute Gravity measurements in the Antarctic region. Maitri is marked by red rectangle. Abbreviations of the names are: NOVO:Novolazarevskaya; DUMO:Dumontd'Urville; TNB:TerraNovaBay; ROBE:CapeRoberts; COAT:Mt.Coates; MCMU:McMurdo; JUBA:Jubany; OHIG: O'Higgins.

METHOD OF MEASURING ABSOLUTE GRAVITY

The only commercially available absolute gravity meter (AGM) is FG5. FG-5 absolute gravimeter is based on measuring time and distance precisely of free falling object inside a vacuumed chamber. Figure 2 is a schematic diagram of the instrument. The decent of the freely-falling object inside a

vacuum chamber (called dropping chamber) is monitored very accurately using a laser interferometer. The optical fringes generated in the interferometer provide a very accurate distance measurement system that can be traced to absolute wavelength standards. Very accurate and precise timing of the occurrence of these optical fringes is done using an atomic rubidium clock. The time and distance data are used in the equation of motion to solve for the acceleration due to gravity ' g '. The whole system is automatically controlled by a computer and typically 100 drops are recorded for one set of gravity observation. Detail procedure of measurements and applications of absolute gravity measurements are documented in the Tiwari et al. (2006).

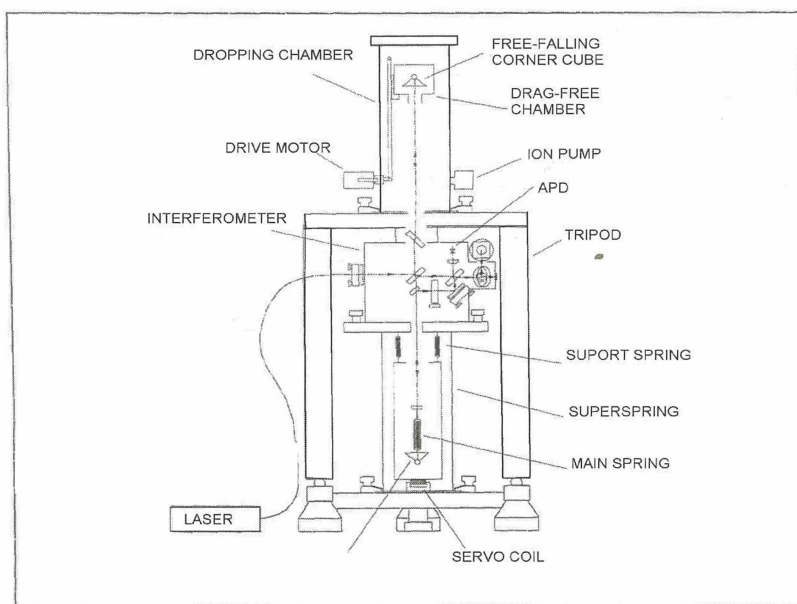


Fig. 2: Schematic diagram of Absolute Gravity Meter FG5

MEASUREMENTS AT MAITRI, ANTARCTICA

We reached Maitri, an Indian Antarctic station on 1st of January 2004 for Absolute Gravity (AG) measurements and a few repeat gravity measurements on the earlier sites. A concrete platform for AG measurements was constructed and covered under Aravali hut with required sealing by

28th January. Before putting the hut, GPS observation of about four hours were made on the reference point of pillar for precise height. Actual observations took off on 4th February, following construction of a good platform in a temperature controlled hut. Before measurements stability of all the components in general and super-spring and laser source, in particular are tested independently. A required temperature (15-25°C) was maintained by auto cut off heater during measurements. Figure 3 is a photograph of the measuring system in the operation at Maitri, Antarctica.



Fig. 3: Aravali hut that housed FG5 Absolute Gravity Meter

We continued the measurements till 27th February, 2004 except during very bad weather days. We have also connected earlier gravity observations with AG reference point. We followed the same observational procedure mentioned above and recorded about sets of observations of 100 drops. The number of effective data is $365\text{sets} \times 100\text{ drops} \times 700 = 25550000$ and provide the absolute gravity at reference pillar $982578797 \mu\text{Gal}$ with an uncertainty of $\sim 7 \mu\text{Gal}$. A typical plot of absolute gravity time series is shown in Figure 4.

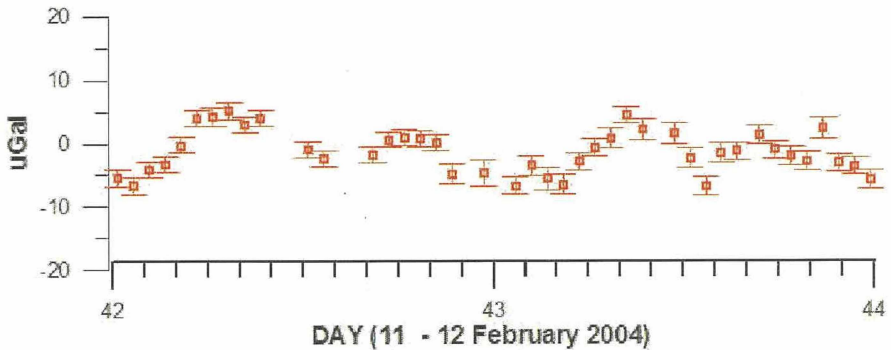


Fig. 4: A time series of absolute gravity value after removing the mean value of 982578797 μGal

We propose to make repeat measurements of established absolute gravity station for temporal gravity changes. Monitoring of temporal variation of gravity field can constrain elastic rebound due to change in ice mass because any change in the ice will deform the earth, with results that could be evident in observations of crustal displacements or of surface gravity (Wahr et al, 2000). In addition to our proposed plan we made a magnetic traverse cutting all the exposed litho units of Schirmacher area. While returning from Antarctica, absolute gravity observations are also made at Parl gravity base in Cape Town in South Africa.

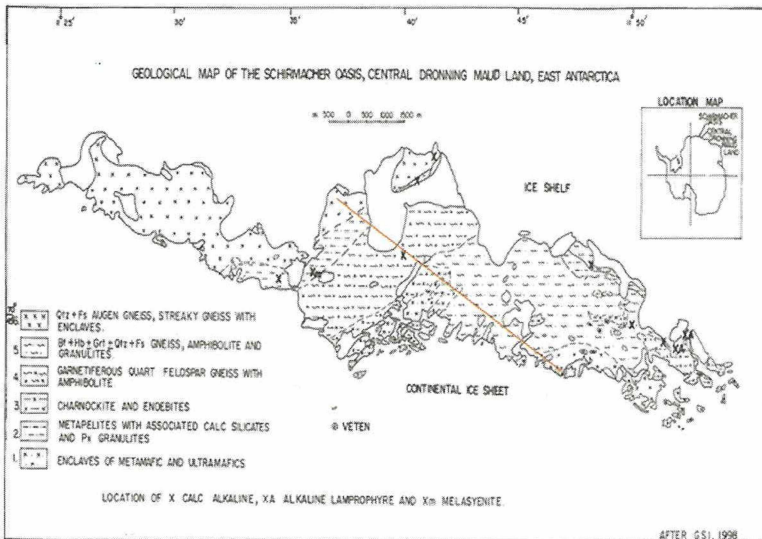


Fig. 5a: Location of traverse along which magnetic data are recorded

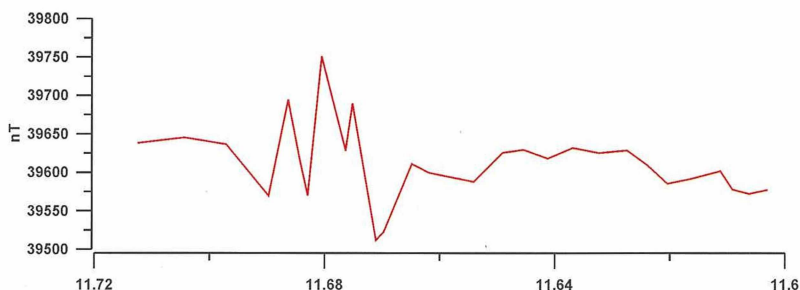


Fig. 5 b : Magnetic profile along the traverse marked in Figure 5a

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